$$C_{ledq}^{i1\_i2\_i3\_i4\_} \to$$

$$\frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} + \frac{1}{16\pi^2} \Big(\frac{1}{24} \frac{1}{m_\Phi^2} s_\gamma^2 \Big(\overline{y}_d^{i4i3} \Big(9 \overline{y}_e^{pr} y_e^{pi2} y_e^{i1r} (1 + s_\gamma^2) + 2 y_e^{i1i2} \Big(11 g_1^2 +$$

$$9 g_2^2 + 24 g_3^2 - 6 c_\gamma^2 (3 \overline{y}_d^{pr} y_d^{pr} + \overline{y}_e^{pr} y_e^{pr}) \Big) \Big) +$$

$$3 y_e^{i1i2} \Big(3 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} (1 + s_\gamma^2) - \overline{y}_d^{pi3} \overline{y}_u^{i4r} y_u^{pr} (-1 + c_\gamma^2) \Big) \Big) +$$

$$\frac{1}{2} \sum_p s_\gamma g_1^2 \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) LF_{1,0}[m_{\tilde{d}}^p] +$$

$$3 s_\gamma \frac{1}{m_\Phi^4} \overline{y}_d^{pr} \overline{y}_d^{i4i3} y_d^{pr} y_e^{i1i2} (-s_{2\gamma} c_\gamma + s_\gamma^3) LF_{1,0}[m_{\tilde{d}}^r] +$$

$$\frac{1}{2} \sum_p s_\gamma g_1^2 \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) LF_{1,0}[m_{\tilde{e}}^p] +$$

$$s_\gamma \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} \overline{y}_e^{pr} y_e^{pr} y_e^{i1i2} (-s_{2\gamma} c_\gamma + s_\gamma^3) LF_{1,0}[m_{\tilde{e}}^r] +$$

$$\frac{1}{2} s_\gamma \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2} \Big(2 \overline{y}_e^{pr} y_e^{pr} (-s_{2\gamma} c_\gamma + s_\gamma^3) - \sum_p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \Big) LF_{1,0}[m_{\tilde{l}}^p] +$$

$$\frac{1}{2} s_\gamma \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2}$$

$$\Big(6 \overline{y}_d^{pr} y_d^{pr} (-s_{2\gamma} c_\gamma + s_\gamma^3) + 6 c_\gamma \overline{y}_u^{pr} y_u^{pr} (s_{2\gamma} + s_\gamma c_\gamma) + \sum_p g_1^2 (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) \Big)$$

$$LF_{1,0}[m_{\tilde{q}}^p] - \sum_p s_\gamma g_1^2 \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2} (2 s_{2\gamma} c_\gamma + s_\gamma c_{2\gamma}) LF_{1,0}[m_{\tilde{u}}^p] +$$

$$3 s_\gamma c_\gamma \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2} \overline{y}_u^{pr} y_u^{pr} (s_{2\gamma} + s_\gamma c_\gamma) LF_{1,0}[m_{\tilde{u}}^r] + \frac{1}{4} s_\gamma \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2}$$

$$\Big(3 s_{4\gamma} c_\gamma (g_1^2 + g_2^2) + s_\gamma \big(g_1^2 (-1 + 3 c_{2\gamma}^2) + 3 g_2^2 (-1 + c_{2\gamma}^2) \big) \Big) LF_{1,0}[m_\Phi] +$$

$$\frac{1}{4} \frac{1}{m_\Phi^2} s_\gamma^2 \Big(\overline{y}_d^{i4i3} \big(-3 s_\gamma^2 \overline{y}_e^{pr} y_e^{pi2} y_e^{i1r} + 2 y_e^{i1i2} (g_1^2 + 3 g_2^2) \big) +$$

$$3 y_e^{i1i2} \big(-s_\gamma^2 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} + c_\gamma^2 \overline{y}_d^{pi3} \overline{y}_u^{i4r} y_u^{pr} \big) \Big) LF_{1,1}[m_\Phi] -$$

$$\frac{1}{4} s_\gamma^2 y_e^{i1i2} \big(\overline{y}_d^{i4i3} (g_1^2 + 3 g_2^2) + 2 c_\gamma^2 \overline{y}_d^{pi3} \overline{y}_u^{i4r} y_u^{pr} \big) LF_{1,2}[m_\Phi] -$$

$$\frac{1}{9} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,0}[m_1, m_{\tilde{d}}^{i3}] + \frac{1}{18} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,1,-1}[m_1, m_{\tilde{d}}^{i3}] -$$

$$g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,0}[m_1, m_{\tilde{e}}^{i2}] + \frac{1}{2} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,1,-1}[m_1, m_{\tilde{e}}^{i2}] -$$

$$\frac{1}{4} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,0}[m_1, m_{\tilde{l}}^{i1}] + \frac{1}{8} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,1,-1}[m_1, m_{\tilde{l}}^{i1}] -$$

$$\frac{1}{36} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,0}[m_1, m_{\tilde{q}}^{i4}] +$$

$$\frac{1}{72} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,1,-1}[m_1, m_{\tilde{q}}^{i4}] - g_1^2 \frac{1}{m_\Phi^4} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,-1}[m_1, \tilde{\mu}] -$$

$$2 m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma^2 - 2 s_\gamma^2) LF_{1,1,0}[m_1, \tilde{\mu}] -$$

$$\frac{3}{4} g_2^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,0}[m_2, m_{\tilde{l}}^{i1}] +$$

$$\frac{3}{8} g_2^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,1,-1}[m_2, m_{\tilde{l}}^{i1}] - \frac{3}{4} g_2^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,0}[m_2, m_{\tilde{q}}^{i4}] +$$

$$\frac{3}{8} g_2^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,1,-1}[m_2, m_{\tilde{q}}^{i4}] - 3 g_2^2 \frac{1}{m_\Phi^4} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,-1}[m_2, \tilde{\mu}] -$$

$$6 m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma^2 - 2 s_\gamma^2) LF_{1,1,0}[m_2, \tilde{\mu}] -$$

$$\frac{4}{3} g_3^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,0}[m_3, m_{\tilde{d}}^{i3}] + \frac{2}{3} g_3^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,1,-1}[m_3, m_{\tilde{d}}^{i3}] -$$

$$\frac{4}{3} g_3^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,0}[m_3, m_{\tilde{q}}^{i4}] + \frac{2}{3} g_3^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,1,-1}[m_3, m_{\tilde{q}}^{i4}] -$$

$$\frac{1}{2} \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} y_e^{i1i2} LF_{1,1,0}[m_{\tilde{d}}^p, \tilde{\mu}] + 3 s_\gamma \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2}$$

$$\Big(s_\gamma \overline{a}_d^{pr} a_d^{pr} (-2 c_\gamma^2 + s_\gamma^2) - \tilde{\mu} c_\gamma (c_\gamma^2 - 2 s_\gamma^2) (\overline{y}_d^{pr} a_d^{pr} + y_d^{pr} \overline{a}_d^{pr}) + 3 s_\gamma \tilde{\mu}^2 c_\gamma^2 \overline{y}_d^{pr} y_d^{pr} \Big)$$

$$LF_{1,1,0}[m_{\tilde{d}}^r, m_{\tilde{q}}^p] + s_\gamma \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2}$$

$$\Big(s_\gamma \overline{a}_e^{pr} a_e^{pr} (-2 c_\gamma^2 + s_\gamma^2) - \tilde{\mu} c_\gamma (c_\gamma^2 - 2 s_\gamma^2) (\overline{y}_e^{pr} a_e^{pr} + y_e^{pr} \overline{a}_e^{pr}) + 3 s_\gamma \tilde{\mu}^2 c_\gamma^2 \overline{y}_e^{pr} y_e^{pr} \Big)$$

$$LF_{1,1,0}[m_{\tilde{e}}^r, m_{\tilde{l}}^p] - \frac{1}{2} \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} \overline{y}_e^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,0}[m_{\tilde{e}}^r, \tilde{\mu}] + s_\gamma c_\gamma \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3}$$

$$y_e^{i1i2} \Big(\tilde{\mu} \overline{y}_e^{pr} \big(a_e^{pr} (c_\gamma^2 - s_\gamma^2) - 2 s_\gamma \tilde{\mu} c_\gamma y_e^{pr} \big) + \overline{a}_e^{pr} \big(2 s_\gamma c_\gamma a_e^{pr} + \tilde{\mu} y_e^{pr} (c_\gamma^2 - s_\gamma^2) \big) \Big)$$

$$LF_{2,1,0}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] + s_\gamma c_\gamma \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} \Big(\tilde{\mu} \overline{y}_e^{pr} \big(a_e^{pr} (-c_\gamma^2 + s_\gamma^2) + 2 s_\gamma \tilde{\mu} c_\gamma y_e^{pr} \big) +$$

$$\overline{a}_e^{pr} \big(-2 s_\gamma c_\gamma a_e^{pr} + \tilde{\mu} y_e^{pr} (-c_\gamma^2 + s_\gamma^2) \big) \Big) LF_{3,1,-1}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] +$$

$$c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma \overline{a}_e^{pr} - s_\gamma \tilde{\mu} \overline{y}_e^{pr}) (c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}) LF_{3,1,0}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] -$$

$$3 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma \overline{a}_e^{pr} - s_\gamma \tilde{\mu} \overline{y}_e^{pr}) (c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}) LF_{4,1,-1}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] +$$

$$2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma \overline{a}_e^{pr} - s_\gamma \tilde{\mu} \overline{y}_e^{pr}) (c_\gamma a_e^{pr} - s_\gamma \tilde{\mu} y_e^{pr}) LF_{5,1,-2}[m_{\tilde{l}}^p, m_{\tilde{e}}^r] -$$

$$\frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} \overline{y}_e^{pr} y_e^{pi2} y_e^{i1r} LF_{1,1,0}[m_{\tilde{l}}^p, \tilde{\mu}] + 3 s_\gamma c_\gamma \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2}$$

$$\Big(\tilde{\mu} \overline{y}_d^{pr} \big(a_d^{pr} (c_\gamma^2 - s_\gamma^2) - 2 s_\gamma \tilde{\mu} c_\gamma y_d^{pr} \big) + \overline{a}_d^{pr} \big(2 s_\gamma c_\gamma a_d^{pr} + \tilde{\mu} y_d^{pr} (c_\gamma^2 - s_\gamma^2) \big) \Big)$$

$$LF_{2,1,0}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] + 3 s_\gamma c_\gamma \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} \Big(\tilde{\mu} \overline{y}_d^{pr} \big(a_d^{pr} (-c_\gamma^2 + s_\gamma^2) + 2 s_\gamma \tilde{\mu} c_\gamma y_d^{pr} \big) +$$

$$\overline{a}_d^{pr} \big(-2 s_\gamma c_\gamma a_d^{pr} + \tilde{\mu} y_d^{pr} (-c_\gamma^2 + s_\gamma^2) \big) \Big) LF_{3,1,-1}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] +$$

$$3 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma \overline{a}_d^{pr} - s_\gamma \tilde{\mu} \overline{y}_d^{pr}) (c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}) LF_{3,1,0}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] -$$

$$9 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma \overline{a}_d^{pr} - s_\gamma \tilde{\mu} \overline{y}_d^{pr}) (c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}) LF_{4,1,-1}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] + 6 c_\gamma^2 \overline{y}_d^{i4i3}$$

$$y_e^{i1i2} (c_\gamma \overline{a}_d^{pr} - s_\gamma \tilde{\mu} \overline{y}_d^{pr}) (c_\gamma a_d^{pr} - s_\gamma \tilde{\mu} y_d^{pr}) LF_{5,1,-2}[m_{\tilde{q}}^p, m_{\tilde{d}}^r] + 3 s_\gamma \frac{1}{m_\Phi^4} \overline{y}_d^{i4i3} y_e^{i1i2}$$

$$\Big(3 s_\gamma c_\gamma^2 \overline{a}_u^{pr} a_u^{pr} - \tilde{\mu} c_\gamma (c_\gamma^2 - 2 s_\gamma^2) (\overline{y}_u^{pr} a_u^{pr} + y_u^{pr} \overline{a}_u^{pr}) + s_\gamma \tilde{\mu}^2 \overline{y}_u^{pr} y_u^{pr} (-2 c_\gamma^2 + s_\gamma^2) \Big)$$

$$LF_{1,1,0}[m_{\tilde{q}}^p, m_{\tilde{u}}^r] - \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} y_e^{i1i2} LF_{1,1,0}[m_{\tilde{q}}^r, \tilde{\mu}] + 3 s_\gamma c_\gamma \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3}$$

$$y_e^{i1i2} \Big(\tilde{\mu} \overline{y}_u^{pr} \big(a_u^{pr} (c_\gamma^2 - s_\gamma^2) + 2 s_\gamma \tilde{\mu} c_\gamma y_u^{pr} \big) + \overline{a}_u^{pr} \big(-2 s_\gamma c_\gamma a_u^{pr} + \tilde{\mu} y_u^{pr} (c_\gamma^2 - s_\gamma^2) \big) \Big)$$

$$LF_{2,1,0}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] + 3 s_\gamma c_\gamma \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} \Big(\tilde{\mu} \overline{y}_u^{pr} \big(a_u^{pr} (-c_\gamma^2 + s_\gamma^2) - 2 s_\gamma \tilde{\mu} c_\gamma y_u^{pr} \big) +$$

$$\overline{a}_u^{pr} \big(2 s_\gamma c_\gamma a_u^{pr} + \tilde{\mu} y_u^{pr} (-c_\gamma^2 + s_\gamma^2) \big) \Big) LF_{3,1,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] +$$

$$3 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} (-s_\gamma \overline{a}_u^{pr} + \tilde{\mu} c_\gamma \overline{y}_u^{pr}) (-s_\gamma a_u^{pr} + \tilde{\mu} c_\gamma y_u^{pr}) LF_{3,1,0}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] -$$

$$9 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} (-s_\gamma \overline{a}_u^{pr} + \tilde{\mu} c_\gamma \overline{y}_u^{pr}) (-s_\gamma a_u^{pr} + \tilde{\mu} c_\gamma y_u^{pr}) LF_{4,1,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] +$$

$$6 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} (-s_\gamma \overline{a}_u^{pr} + \tilde{\mu} c_\gamma \overline{y}_u^{pr}) (-s_\gamma a_u^{pr} + \tilde{\mu} c_\gamma y_u^{pr}) LF_{5,1,-2}[m_{\tilde{u}}^r, m_{\tilde{q}}^p] -$$

$$\frac{1}{2} \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} y_u^{pr} LF_{1,1,0}[m_{\tilde{u}}^r, \tilde{\mu}] +$$

$$2 m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma^2 - s_\gamma^2) LF_{2,1,0}[\tilde{\mu}, m_1] -$$

$$2 c_\gamma g_1^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} \big(c_\gamma m_\Phi^2 + m_1 s_\gamma \tilde{\mu} (c_\gamma^2 - s_\gamma^2) \big) LF_{3,1,-1}[\tilde{\mu}, m_1] -$$

$$2 m_1 s_\gamma \tilde{\mu} g_1^2 c_\gamma^3 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{3,1,0}[\tilde{\mu}, m_1] + 4 g_1^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{4,1,-2}[\tilde{\mu}, m_1] +$$

$$6 m_1 s_\gamma \tilde{\mu} g_1^2 c_\gamma^3 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{4,1,-1}[\tilde{\mu}, m_1] -$$

$$2 g_1^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{5,1,-3}[\tilde{\mu}, m_1] - 4 m_1 s_\gamma \tilde{\mu} g_1^2 c_\gamma^3 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{5,1,-2}[\tilde{\mu}, m_1] +$$

$$6 m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} (c_\gamma^2 - s_\gamma^2) LF_{2,1,0}[\tilde{\mu}, m_2] -$$

$$6 c_\gamma g_2^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} \big(c_\gamma m_\Phi^2 + m_2 s_\gamma \tilde{\mu} (c_\gamma^2 - s_\gamma^2) \big) LF_{3,1,-1}[\tilde{\mu}, m_2] -$$

$$6 m_2 s_\gamma \tilde{\mu} g_2^2 c_\gamma^3 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{3,1,0}[\tilde{\mu}, m_2] + 12 g_2^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{4,1,-2}[\tilde{\mu}, m_2] +$$

$$18 m_2 s_\gamma \tilde{\mu} g_2^2 c_\gamma^3 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{4,1,-1}[\tilde{\mu}, m_2] -$$

$$6 g_2^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{5,1,-3}[\tilde{\mu}, m_2] - 12 m_2 s_\gamma \tilde{\mu} g_2^2 c_\gamma^3 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{5,1,-2}[\tilde{\mu}, m_2] +$$

$$\frac{1}{4} \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} y_e^{i1i2} LF_{2,1,-1}[\tilde{\mu}, m_{\tilde{d}}^p] +$$

$$\frac{1}{4} \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} \overline{y}_e^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,-1}[\tilde{\mu}, m_{\tilde{e}}^r] +$$

$$\frac{1}{2} \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} \overline{y}_e^{pr} y_e^{pi2} y_e^{i1r} LF_{2,1,-1}[\tilde{\mu}, m_{\tilde{l}}^p] +$$

$$\frac{1}{2} \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{ri3} \overline{y}_d^{i4p} y_d^{rp} y_e^{i1i2} LF_{2,1,-1}[\tilde{\mu}, m_{\tilde{q}}^r] +$$

$$\frac{1}{4} \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} y_u^{pr} LF_{2,1,-1}[\tilde{\mu}, m_{\tilde{u}}^r] -$$

$$\frac{1}{9} m_1 s_\gamma g_1^2 \frac{1}{m_\Phi^2} y_e^{i1i2} (s_\gamma \overline{a}_d^{i4i3} + \tilde{\mu} c_\gamma \overline{y}_d^{i4i3}) LF_{1,1,1,0}[m_1, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i4}] +$$

$$\frac{1}{36} m_1 c_\gamma g_1^2 y_e^{i1i2} (c_\gamma \overline{a}_d^{i4i3} - s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{2,2,1,-1}[m_1, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i4}] +$$

$$\frac{1}{3} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,-1}[m_1, m_{\tilde{d}}^{i3}, \tilde{\mu}] -$$

$$\frac{1}{3} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,0}[m_1, m_{\tilde{d}}^{i3}, \tilde{\mu}] +$$

$$m_1 s_\gamma g_1^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} (s_\gamma a_e^{i1i2} + \tilde{\mu} c_\gamma y_e^{i1i2}) LF_{1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] +$$

$$\frac{1}{4} m_1 c_\gamma g_1^2 \overline{y}_d^{i4i3} (-c_\gamma a_e^{i1i2} + s_\gamma \tilde{\mu} y_e^{i1i2}) LF_{2,2,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{l}}^{i1}] +$$

$$g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] -$$

$$m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,0}[m_1, m_{\tilde{e}}^{i2}, \tilde{\mu}] +$$

$$\frac{1}{4} m_1 c_\gamma g_1^2 \overline{y}_d^{i4i3} (-c_\gamma a_e^{i1i2} + s_\gamma \tilde{\mu} y_e^{i1i2}) LF_{2,2,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{e}}^{i2}] -$$

$$\frac{1}{2} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$

$$\frac{1}{2} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,0}[m_1, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$

$$\frac{1}{36} m_1 c_\gamma g_1^2 y_e^{i1i2} (c_\gamma \overline{a}_d^{i4i3} - s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{2,2,1,-1}[m_1, m_{\tilde{q}}^{i4}, m_{\tilde{d}}^{i3}] +$$

$$\frac{1}{6} g_1^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,-1}[m_1, m_{\tilde{q}}^{i4}, \tilde{\mu}] -$$

$$\frac{1}{6} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,0}[m_1, m_{\tilde{q}}^{i4}, \tilde{\mu}] +$$

$$\frac{1}{6} g_1^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{d}}^{i3}] +$$

$$\frac{1}{6} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{d}}^{i3}] +$$

$$\frac{1}{2} g_1^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] +$$

$$\frac{1}{2} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{e}}^{i2}] -$$

$$\frac{1}{4} g_1^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] -$$

$$\frac{1}{4} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{l}}^{i1}] +$$

$$\frac{1}{12} g_1^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i4}] +$$

$$\frac{1}{12} m_1 s_\gamma \tilde{\mu} c_\gamma g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-1}[m_1, \tilde{\mu}, m_{\tilde{q}}^{i4}] +$$

$$\frac{3}{2} g_2^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] -$$

$$\frac{3}{2} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,0}[m_2, m_{\tilde{l}}^{i1}, \tilde{\mu}] +$$

$$\frac{3}{2} g_2^2 \frac{1}{m_\Phi^2} s_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,-1}[m_2, m_{\tilde{q}}^{i4}, \tilde{\mu}] -$$

$$\frac{3}{2} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \frac{1}{m_\Phi^2} \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,0}[m_2, m_{\tilde{q}}^{i4}, \tilde{\mu}] +$$

$$\frac{3}{4} g_2^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] +$$

$$\frac{3}{4} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{l}}^{i1}] +$$

$$\frac{3}{4} g_2^2 c_\gamma^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i4}] +$$

$$\frac{3}{4} m_2 s_\gamma \tilde{\mu} c_\gamma g_2^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{2,2,1,-1}[m_2, \tilde{\mu}, m_{\tilde{q}}^{i4}] +$$

$$\frac{8}{3} m_3 s_\gamma g_3^2 \frac{1}{m_\Phi^2} y_e^{i1i2} (s_\gamma \overline{a}_d^{i4i3} + \tilde{\mu} c_\gamma \overline{y}_d^{i4i3}) LF_{1,1,1,0}[m_3, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i4}] +$$

$$\frac{2}{3} m_3 c_\gamma g_3^2 y_e^{i1i2} (-c_\gamma \overline{a}_d^{i4i3} + s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{2,2,1,-1}[m_3, m_{\tilde{d}}^{i3}, m_{\tilde{q}}^{i4}] +$$

$$\frac{2}{3} m_3 c_\gamma g_3^2 y_e^{i1i2} (-c_\gamma \overline{a}_d^{i4i3} + s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{2,2,1,-1}[m_3, m_{\tilde{q}}^{i4}, m_{\tilde{d}}^{i3}] +$$

$$\frac{1}{18} m_1 c_\gamma g_1^2 y_e^{i1i2} (-c_\gamma \overline{a}_d^{i4i3} + s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{2,1,1,0}[m_{\tilde{d}}^{i3}, m_1, m_{\tilde{q}}^{i4}] +$$

$$\frac{1}{18} m_1 c_\gamma g_1^2 y_e^{i1i2} (c_\gamma \overline{a}_d^{i4i3} - s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{3,1,1,-1}[m_{\tilde{d}}^{i3}, m_1, m_{\tilde{q}}^{i4}] +$$

$$\frac{4}{3} m_3 c_\gamma g_3^2 y_e^{i1i2} (c_\gamma \overline{a}_d^{i4i3} - s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{2,1,1,0}[m_{\tilde{d}}^{i3}, m_3, m_{\tilde{q}}^{i4}] +$$

$$\frac{4}{3} m_3 c_\gamma g_3^2 y_e^{i1i2} (-c_\gamma \overline{a}_d^{i4i3} + s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{3,1,1,-1}[m_{\tilde{d}}^{i3}, m_3, m_{\tilde{q}}^{i4}] +$$

$$\frac{1}{2} m_1 c_\gamma g_1^2 \overline{y}_d^{i4i3} (c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}) LF_{2,1,1,0}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] +$$

$$\frac{1}{2} m_1 c_\gamma g_1^2 \overline{y}_d^{i4i3} (-c_\gamma a_e^{i1i2} + s_\gamma \tilde{\mu} y_e^{i1i2}) LF_{3,1,1,-1}[m_{\tilde{e}}^{i2}, m_1, m_{\tilde{l}}^{i1}] +$$

$$\frac{1}{2} m_1 c_\gamma g_1^2 \overline{y}_d^{i4i3} (c_\gamma a_e^{i1i2} - s_\gamma \tilde{\mu} y_e^{i1i2}) LF_{2,1,1,0}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}] +$$

$$\frac{1}{2} m_1 c_\gamma g_1^2 \overline{y}_d^{i4i3} (-c_\gamma a_e^{i1i2} + s_\gamma \tilde{\mu} y_e^{i1i2}) LF_{3,1,1,-1}[m_{\tilde{l}}^{i1}, m_1, m_{\tilde{e}}^{i2}] +$$

$$s_\gamma \tilde{\mu} \frac{1}{m_\Phi^2} \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} (c_\gamma a_u^{pr} + s_\gamma \tilde{\mu} y_u^{pr}) LF_{1,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{u}}^r, \tilde{\mu}] +$$

$$\frac{1}{2} \tilde{\mu} c_\gamma \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} (-s_\gamma a_u^{pr} + \tilde{\mu} c_\gamma y_u^{pr}) LF_{2,1,1,0}[m_{\tilde{q}}^p, m_{\tilde{u}}^r, \tilde{\mu}] +$$

$$\frac{1}{2} \tilde{\mu} c_\gamma \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) LF_{3,1,1,-1}[m_{\tilde{q}}^p, m_{\tilde{u}}^r, \tilde{\mu}] +$$

$$\frac{1}{4} \tilde{\mu} c_\gamma \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) LF_{2,2,1,-1}[m_{\tilde{q}}^p, \tilde{\mu}, m_{\tilde{u}}^r] +$$

$$\frac{1}{18} m_1 c_\gamma g_1^2 y_e^{i1i2} (-c_\gamma \overline{a}_d^{i4i3} + s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{2,1,1,0}[m_{\tilde{q}}^{i4}, m_1, m_{\tilde{d}}^{i3}] +$$

$$\frac{1}{18} m_1 c_\gamma g_1^2 y_e^{i1i2} (c_\gamma \overline{a}_d^{i4i3} - s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{3,1,1,-1}[m_{\tilde{q}}^{i4}, m_1, m_{\tilde{d}}^{i3}] +$$

$$\frac{4}{3} m_3 c_\gamma g_3^2 y_e^{i1i2} (c_\gamma \overline{a}_d^{i4i3} - s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{2,1,1,0}[m_{\tilde{q}}^{i4}, m_3, m_{\tilde{d}}^{i3}] +$$

$$\frac{4}{3} m_3 c_\gamma g_3^2 y_e^{i1i2} (-c_\gamma \overline{a}_d^{i4i3} + s_\gamma \tilde{\mu} \overline{y}_d^{i4i3}) LF_{3,1,1,-1}[m_{\tilde{q}}^{i4}, m_3, m_{\tilde{d}}^{i3}] +$$

$$\frac{1}{2} \tilde{\mu} c_\gamma \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} (-s_\gamma a_u^{pr} + \tilde{\mu} c_\gamma y_u^{pr}) LF_{2,1,1,0}[m_{\tilde{u}}^r, m_{\tilde{q}}^p, \tilde{\mu}] +$$

$$\frac{1}{2} \tilde{\mu} c_\gamma \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) LF_{3,1,1,-1}[m_{\tilde{u}}^r, m_{\tilde{q}}^p, \tilde{\mu}] +$$

$$\frac{1}{4} \tilde{\mu} c_\gamma \overline{y}_d^{pi3} y_e^{i1i2} \overline{y}_u^{i4r} (s_\gamma a_u^{pr} - \tilde{\mu} c_\gamma y_u^{pr}) LF_{2,2,1,-1}[m_{\tilde{u}}^r, \tilde{\mu}, m_{\tilde{q}}^p] -$$

$$\frac{1}{3} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,1,-1}[m_1, m_{\tilde{d}}^{i3}, m_{\tilde{e}}^{i2}, \tilde{\mu}] +$$

$$\frac{1}{6} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,1,-1}[m_1, m_{\tilde{d}}^{i3}, m_{\tilde{l}}^{i1}, \tilde{\mu}] -$$

$$\frac{1}{6} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,1,-1}[m_1, m_{\tilde{e}}^{i2}, m_{\tilde{q}}^{i4}, \tilde{\mu}] +$$

$$\frac{1}{12} g_1^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,1,-1}[m_1, m_{\tilde{l}}^{i1}, m_{\tilde{q}}^{i4}, \tilde{\mu}] -$$

$$\frac{3}{4} g_2^2 \overline{y}_d^{i4i3} y_e^{i1i2} LF_{1,1,1,1,-1}[m_2, m_{\tilde{l}}^{i1}, m_{\tilde{q}}^{i4}, \tilde{\mu}] \Big)$$